

Ulceroglandular Tularemia: A Comprehensive Disease Characteristics Report

Disease: Ulceroglandular Tularemia **MONDO ID:** MONDO:0001413 **Category:** Infectious Disease (zoonotic, bacterial) **Causative agent:** *Francisella tularensis* (NCBITaxon:263) **Evidence base:** Literature-derived (aggregated disease-level resources, systematic reviews, CDC surveillance, and mechanistic studies in cell/mouse models). No individual patient (EHR) data were used.

Summary

Ulceroglandular tularemia is the most common clinical form of tularemia, an acute zoonotic infection caused by *Francisella tularensis*, a small, aerobic, Gram-negative, facultative intracellular coccobacillus and one of the most infectious bacterial pathogens known (infectious dose as low as ~10 organisms). The ulceroglandular form arises specifically from **cutaneous inoculation** of the organism — most often through the bite of an infected arthropod (ticks such as *Dermacentor variabilis* and *Amblyomma americanum*, deer flies, and mosquitoes) or through direct handling of infected animals (lagomorphs and rodents). After a 3–6 day incubation period, patients develop an acute febrile illness with a papule that ulcerates at the site of inoculation (skin ulcer, HP:0200042) accompanied by painful regional lymphadenopathy (HP:0002716). This is fundamentally an **infectious, not a genetic, disease**: there are no causal human genes, pathogenic variants, or heritable susceptibility loci, so the "genetic/molecular" content of this report concerns the *bacterial* virulence determinants rather than host germline genetics.

The pathophysiology is driven by the bacterium's ability to survive and replicate inside host phagocytes. The **Francisella Pathogenicity Island (FPI)**, which encodes a **Type VI secretion system (T6SS)**, enables the bacterium to escape the phagosome and replicate freely in the macrophage cytosol, ultimately triggering inflammasome activation and host-cell death. This intracellular lifestyle explains why host defense is predominantly **cell-mediated** (T-cell/IFN- γ

dependent): granulocytes cannot kill the organism without opsonizing antibody, and protective immunity emerges as a vigorous T-lymphocyte response 1–2 weeks after infection, with humoral antibody appearing at 2–3 weeks and serving primarily a diagnostic role.

Clinically, ulceroglandular tularemia has a **good prognosis** because the pathogen is well contained by a vigorous local inflammatory reaction, contrasting with the typhoidal syndrome, which has fewer localizing signs, more pneumonia, and higher untreated mortality. Diagnosis relies on serology (microagglutination titer $\geq 1/160$ or a four-fold rise) and PCR of ulcer swabs or lymph-node aspirates; culture is hazardous and requires BSL-3 containment. Treatment with aminoglycosides (streptomycin, gentamicin — first-line), fluoroquinolones (ciprofloxacin), or tetracyclines (doxycycline) yields case-fatality rates below ~1.2%. There is **no licensed vaccine**; prevention rests on exposure avoidance and, after high-risk exposures, post-exposure antibiotic prophylaxis. *F. tularensis* is classified as a **Tier-1 Select Agent / Category A bioterrorism agent** owing to its low infectious dose and aerosol infectivity.

Section 1 — Disease Information

Overview. Ulceroglandular tularemia is the cutaneous-inoculation form of tularemia, a zoonosis caused by *Francisella tularensis*. Across a systematic review of 870 cases spanning 1993–2023 in 35 countries, **ulceroglandular disease was the single most common clinical form**, followed by oropharyngeal, glandular, and pneumonic disease — *"The most common clinical forms were ulceroglandular, oropharyngeal, glandular, and pneumonic disease"* (PMID: 38294108). It is defined by the route of infection — cutaneous inoculation producing a local ulcer plus regional lymphadenitis (PMID: 32989563).

Key identifiers. - **MONDO:** MONDO:0001413 (ulceroglandular tularemia) - **MeSH:** Tularemia (D014406) - **ICD-10:** A21.0 (Ulceroglandular tularaemia); parent A21 (Tularaemia) - **ICD-11:** 1B94 (Tularaemia) - **SNOMED CT:** Ulceroglandular tularemia (disorder) - **OMIM / Orphanet:** Not applicable as a heritable disorder (infectious disease; no OMIM Mendelian entry).

Synonyms / alternative names. Ulceroglandular tularaemia (British spelling); "rabbit fever," "deer-fly fever," "Ohara disease," "Francis disease," and "Pahvant Valley plague" are historical synonyms for tularemia broadly, of which the ulceroglandular form is the classic presentation.

Information source. The evidence is derived from **aggregated disease-level resources** — systematic reviews, national surveillance (CDC NNDSS), and case series — rather than individual EHR-derived patient records.

Section 2 — Etiology

Primary cause (infectious). The disease is caused entirely by infection with *Francisella tularensis*. There is **no genetic (host) etiology**; heritable variants, susceptibility loci, and modifier genes are **not applicable**. Two clinically important subspecies exist: subsp. *tularensis* (**type A**, most virulent, North America) and subsp. *holarctica* (**type B**, milder, Northern Hemisphere including Europe and Asia) ([PMID: 32989563](#)).

Environmental / exposure risk factors. - **Arthropod bites** — ticks (*Dermacentor variabilis*, *Amblyomma americanum*), deer flies, horseflies, and mosquitoes. "The disease spreads through vectors such as mosquitoes, horseflies, deer flies, and ticks" ([PMID: 32989563](#)); the principal US tick vectors are *D. variabilis* and *A. americanum* ([PMID: 40788927](#)). - **Handling infected animals / carcasses** — "The common route of transmission in Central Europe is handling infected animals" ([PMID: 34990926](#)). - **Occupational / recreational** — hunters, trappers, farmers, landscapers, veterinarians, and laboratory workers. - **Demographic** — highest incidence in children aged 5–9 years and older adult males (bimodal), and markedly elevated among American Indian/Alaska Native persons in the US ([PMID: 39736154](#)).

Protective factors. No genetic protective variants are described (not a host-genetic disease). Environmental protection is behavioral: use of insect repellents, protective clothing, tick checks, and gloves when handling animal carcasses (see Section 13).

Gene–environment interactions. Not applicable in the human-host genetic sense. The relevant "gene–environment" axis is the **bacterial** genotype (type A vs type B; FPI integrity) interacting with the route and dose of exposure to determine clinical form and severity.

Section 3 — Phenotypes

The ulceroglandular syndrome is a localized-plus-regional presentation following a 3–6 day incubation.

Phenotype	Type	HPO term	Frequency / notes
Fever	Symptom / sign	HP:0001945	Near-universal at onset; acute (PMID: 3892222)
Skin ulcer at inoculation site	Physical manifestation	HP:0200042	Defining lesion; papule → ulcer (PMID: 32989563)
Regional lymphadenopathy (painful)	Clinical sign	HP:0002716 / HP:0002840	Defining feature; can suppurate/drain (PMID: 16936340)
Cervical lymphadenopathy (pediatric)	Clinical sign	HP:0002218	Reported after tick bite in young children (PMID: 34990926)
Suppurative lymphadenitis	Complication	—	Especially with delayed therapy (PMID: 16936340)
Secondary skin eruptions (erythema nodosum, Sweet syndrome)	Physical manifestation	HP:0012219 / —	Secondary manifestations in ~15% of tularemia overall; more common in oropharyngeal form (PMID: 29761532)

Characteristics. - **Age of onset:** Any age; the disease is acquired, not congenital. US incidence peaks in children 5–9 and older adults (PMID: 39736154). - **Severity:** Generally mild-to-moderate for the ulceroglandular form (type B predominant in Europe); good prognosis. *"In ulceroglandular tularemia the pathogen appears to be well contained by a vigorous inflammatory reaction. Pneumonia is less common and the patient's prognosis is good"* (PMID: 3892222). - **Progression:** Acute onset; resolving with therapy; lymphadenopathy may persist or suppurate. - **Quality of life:** Acute febrile illness with painful lymphadenopathy causes short-term functional impairment; suppurative nodes may require drainage. Long-term sequelae are uncommon with timely treatment. (No disease-specific EQ-5D/SF-36 data identified — knowledge gap.)

Section 4 — Genetic / Molecular Information

Host genetics: not applicable. Ulceroglandular tularemia is an acquired infection with **no causal human genes, pathogenic germline/somatic variants, modifier genes, epigenetic lesions, or chromosomal abnormalities**. ClinVar/OMIM/HGMD entries do not apply.

Bacterial molecular determinants (the relevant "molecular information"). Virulence depends on the **Francisella Pathogenicity Island (FPI)**, which encodes a **Type VI secretion system (T6SS)**. *"Required for these processes is the Francisella Pathogenicity Island (FPI), which encodes a Type VI secretion system (T6SS) that is active during intracellular infection"* (PMID: 27830989). Key FPI genes include *iglB*, *iglC*, *iglE*, *iglG*, *pdpC*, *dotU*, and *vgrG*. Deletion mutants (Δ iglB, Δ iglE, Δ pdpC) fail to escape the phagosome, do not replicate intracellularly, and are markedly attenuated in the mouse model (PMID: 27830989; PMID: 23356941; PMID: 23403609). PdpC additionally has a regulatory role over *iglABCD* expression, and its deletion abolishes phagosomal escape and cytopathogenicity (PMID: 27477000).

Section 5 — Environmental Information

- **Environmental factors:** The organism persists in the environment (water, soil, animal carcasses) and is maintained in an enzootic cycle. Tularemia is a **Holarctic zoonosis** (PMID: 31600457).
 - **Lifestyle factors:** Outdoor occupational/recreational activities (hunting, trapping, farming, landscaping) increase exposure. *"Humans can acquire the disease through direct contact of sick animals, consumption of infected animals, drinking or direct contact of contaminated water, and inhalation of bacteria-loaded aerosols"* (PMID: 32989563) — the latter routes are more relevant to oropharyngeal/pneumonic forms.
 - **Infectious agent (NCBI Taxonomy):** *Francisella tularensis* (NCBITaxon:263). Subspecies: subsp. *tularensis* (type A) and subsp. *holarctica* (type B, NCBITaxon:264). Type B predominates in Europe and causes the ulceroglandular cases reported from Poland and the Czech Republic (PMID: 34696556; PMID: 32989563).
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Section 6 — Mechanism / Pathophysiology

Causal chain (initiating exposure → clinical manifestation)

1. **Cutaneous inoculation** of *F. tularensis* (~10 organisms) via arthropod bite or contact with an infected animal **leads to** local bacterial deposition in the skin and subcutaneous tissue. *"Francisella tularensis is an extremely virulent pathogen capable of initiating infection with as few as 10 organisms inoculated subcutaneously"* (PMID: 3892222). (Demonstrated.)
2. Local deposition **results in** uptake by resident and recruited **phagocytes** — *"the host responds first with polymorphonuclear leukocytes and then macrophages"* (PMID: 3892222). (Demonstrated.)
3. Inside the phagocyte, the **FPI-encoded T6SS** mediates **escape from the phagosome** into the cytosol. *"Genes in the FPI are required for F. tularensis to escape from the phagosome and replicate in the cytosol"* (PMID: 23403609). (Demonstrated — FPI mutants fail to escape.)
4. Cytosolic localization **leads to rapid intracellular replication** — *"effective intramacrophage proliferation, which is preceded by phagosomal escape into the cytosol"* (PMID: 23356941). (Demonstrated.)
5. Intracellular infection **induces** macrophage **PGE2 synthesis** (dampening adaptive immunity) and **inflammasome activation** (IL-1 β / LDH release). (Demonstrated in macrophage models; PMID: 23403609.)
6. Inflammasome activation and cytosolic burden **result in host-cell death** — *"ultimately causing inflammasome activation and host cell death"* (PMID: 27830989) — releasing bacteria to infect neighboring cells and draining to regional lymph nodes. (Demonstrated.)
7. Bacterial drainage to regional nodes **leads to** a vigorous local **inflammatory / granulomatous response** — the clinical **skin ulcer + regional lymphadenopathy**, sometimes progressing to **suppurative lymphadenitis**. (Inferred from clinical–pathological correlation.)
8. **Branch:** In the ulceroglandular form the pathogen is **well contained** locally → good prognosis. *"In typhoidal disease there are few localizing signs; pneumonia is more common; and the mortality without therapy is much higher, suggesting that the host response is somehow deficient"* (PMID: 3892222). (Inferred from syndrome comparison.)
9. Adaptive control **requires** a **cell-mediated (T-cell/IFN- γ)** response appearing 1–2 weeks post-infection; humoral antibody at 2–3 weeks is largely diagnostic. (Demonstrated.)

Detail by category

- **Cellular processes (GO):** phagocytosis (GO:0006909); phagosomal escape; intracellular replication; inflammasome/pyroptosis (GO:0070269); inflammatory response (GO:0006954).
 - **Immune involvement:** Cell-mediated immunity is dominant. *"Granulocytes are unable to kill the pathogen without opsonizing antibody leaving cellular immunity to play the major role in host defense"* (PMID: 3892222). Both natural infection and LVS vaccination drive **multifunctional T-cell responses** (PMID: 42238599). In models, perforin/granzyme cytotoxicity and NK-cell activity contribute to protection (PMID: 22493083).
 - **Protein dysfunction / secretion:** T6SS structural and effector proteins (IglB, IglE, PdpC, VgrG) mediate the secretion required for escape; N-terminal residues of IglE critically control T6SS-mediated secretion (PMID: 27830989).
 - **Tissue damage mechanisms:** host-cell death (pyroptosis/necrosis), suppuration/abscess formation in lymph nodes, granulomatous inflammation.
 - **Cell types (CL):** macrophage (CL:0000235), monocyte-derived macrophage, neutrophil (CL:0000775), T lymphocyte (CL:0000084), NK cell (CL:0000623).
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Section 7 — Anatomical Structures Affected

- **Primary organs / sites (UBERON):** skin (UBERON:0002097) at the inoculation site — the ulcer; and **regional lymph nodes** (UBERON:0000029) draining that site (e.g., axillary, inguinal, epitrochlear; cervical in children).
- **Secondary involvement:** In disseminated/severe disease, spleen (UBERON:0002106) and liver (UBERON:0002107) can be affected (demonstrated in mouse models with pathological changes in spleen and liver, PMID: 27477000); lungs (UBERON:0002048) in pneumonic progression (less common in ulceroglandular).
- **Body systems:** integumentary and lymphatic/immune systems primarily; reticuloendothelial system secondarily.
- **Tissue / cell level:** epithelial and connective tissue of skin; lymphoid tissue; principal cellular niche is the **macrophage** (CL:0000235).
- **Subcellular level (GO cellular component):** phagosome / *Francisella*-containing phagosome (GO:0045335) and host **cytosol** (GO:0005829) — the replicative compartment.

- **Localization / lateralization:** Unilateral, following the site of the bite/inoculation; lymphadenopathy is typically ipsilateral to the ulcer.

Section 8 — Temporal Development

- **Onset:** Acquired at any age; **incubation 3–6 days** after subcutaneous inoculation (PMID: 3892222). Onset is **acute**.
- **Progression / stages:** (i) local papule → ulcer; (ii) regional painful lymphadenopathy; (iii) possible suppuration/drainage of involved nodes, particularly when antibiotics are started late — *"Late initiation antibiotic therapy could not prevent suppuration and draining of the involved lymph nodes"* (PMID: 16936340).
- **Immune timeline:** T-cell response detectable at 1–2 weeks; antibody at 2–3 weeks (PMID: 3892222).
- **Course / duration:** Self-limiting to resolving with therapy; generally **not** chronic or lifelong. Prognosis good for the ulceroglandular form.
- **Critical intervention window:** Early antibiotic initiation is the key intervention; late therapy cannot always prevent lymph-node suppuration (PMID: 16936340).

Section 9 — Inheritance and Population

- **Inheritance: Not applicable** — infectious, non-heritable. Penetrance, expressivity, anticipation, mosaicism, founder effects, consanguinity, and carrier frequency are all not applicable.
- **Epidemiology (US):** Average annual incidence **0.064/100,000 (2011–2022)**, 56% higher than 2001–2010 (0.041). *"During 2011-2022, a total of 47 states reported 2,462 tularemia cases, but four central states (Arkansas, Kansas, Missouri, and Oklahoma) accounted for 50% of all reported cases"* (PMID: 39736154). During 2001–2010, *"a total of 1,208 cases were reported (median: 126.5 cases per year; range: 90-154)"* (PMID: 24280916). Reported from all US states except Hawaii.
- **Age / sex:** Bimodal — *"Incidence was highest among children aged 5-9 years (0.083 per 100,000 population) and adult males aged 65-84 years (range = 0.133-0.161)"* (PMID: 39736154). Male predominance.

- **Race/ethnicity:** *"Incidence among American Indian or Alaska Native persons (0.260) was approximately five times that among White persons (0.057)"* (PMID: 39736154).
- **Geographic distribution:** Holarctic (Northern Hemisphere). In Europe, subsp. *holarctica* (type B) predominates; the Czech Republic reports up to ~100 cases/yr (225 and 222 in the 1998–1999 epidemic years) (PMID: 32989563). Ulceroglandular is the most common form reported from Poland (PMID: 34696556); oropharyngeal predominates in Turkey (water-borne outbreaks).

Section 10 — Diagnostics

- **Serology (mainstay):** microagglutination test (MAT), titer $\geq 1/160$ positive — *"microagglutination test (MAT) was performed for all patients whose clinical symptoms were consistent with tularemia and MAT titers $\geq 1/160$ were considered positive"* (PMID: 29642835); some centers use $\geq 1:640$ in outbreak settings (PMID: 21341160). A four-fold rise between acute and convalescent sera is confirmatory. Antibody appears at 2–3 weeks.
- **PCR / NAAT:** Positive on **lymph-node aspiration material** and ulcer swabs — *"PCR for *F. tularensis* was positive in aspiration material of suppurated lymphadenitis of 7 patients"* (PMID: 16936340). RD1-primer PCR can subtype to subsp. *holarctica* (PMID: 21341160).
- **Direct fluorescent antibody (DFA):** Applicable to lymph aspirates.
- **Culture:** Definitive but **hazardous** — requires **BSL-3** containment; low sensitivity from routine specimens (cultures often negative) (PMID: 21341160).
- **Routine labs:** Nonspecific (mild WBC changes, elevated ESR/CRP) (PMID: 21341160).
- **Genetic / omics testing:** Not applicable for host diagnosis (no human genetic testing, karyotyping, CMA, or repeat-expansion testing). Pathogen genomics is used for subtyping/epidemiology.
- **Differential diagnosis:** cat-scratch disease, pyogenic bacterial lymphadenitis, plague, cutaneous anthrax, sporotrichosis, atypical mycobacterial infection, syphilitic chancre, and (in the neck) tonsillopharyngitis / neck mass of unknown origin (PMID: 34696556; PMID: 21341160). Diagnosis is often **delayed** because of low prevalence and nonspecific features.

Section 11 — Outcome / Prognosis

- **Prognosis: Good** for the ulceroglandular form — the pathogen is well contained by a vigorous inflammatory reaction, pneumonia is less common, and outcomes are favorable (PMID: 3892222).
- **Case fatality:** Low with treatment — *"Among patients treated with aminoglycosides (n = 452 [52%]), fluoroquinolones (n = 339 [39%]), or tetracyclines (n = 419 [48%]), the fatality rate was 0.7%, 0.9%, and 1.2%, respectively"* (PMID: 38294108). US surveillance case-fatality is typically <2%.
- **Complications:** Suppurative lymphadenitis with drainage (favored by late therapy) (PMID: 16936340); rarely dissemination.
- **Prognostic factors: Timeliness of antibiotic initiation** is the key modifiable factor; clinical form (ulceroglandular better than typhoidal/pneumonic) and subspecies (type B milder than type A) shape outcome.

Section 12 — Treatment

Drug class	Examples	Role	Fatality in review	NCIT
Aminoglycosides	Streptomycin, gentamicin	First-line / treatment of choice	0.7% (n=452)	NCIT:C255 (Aminoglycoside)
Fluoroquinolones	Ciprofloxacin	Effective alternative / PEP	0.9% (n=339)	NCIT:C1728 (Ciprofloxacin)
Tetracyclines	Doxycycline	Alternative / PEP	1.2% (n=419)	NCIT:C641 (Doxycycline)

- **First-line:** Streptomycin or gentamicin (aminoglycosides) are the treatment of choice (PMID: 29183485; PMID: 41026652). *"Quinolones, tetracyclines, or aminoglycosides are frequently used in the treatment of tularemia"* (PMID: 32989563).
- **Timing matters:** Late initiation cannot always prevent lymph-node suppuration and drainage (PMID: 16936340).

- **Surgical/interventional:** Incision and drainage or excision of suppurative nodes may be needed as adjunct.
- **Advanced/targeted/gene/cell/RNA therapies:** Not applicable.
- **Pharmacogenomics:** Not applicable (no host-genotype-guided dosing established for this indication).
- **Treatment algorithm:** Empiric aminoglycoside for confirmed/strongly suspected disease; doxycycline or ciprofloxacin for milder disease or when aminoglycosides are contraindicated; drainage for suppurative nodes; CDC 2025 guidance governs both naturally acquired and bioterrorism-response treatment ([PMID: 41026652](#)).

Section 13 — Prevention

- **Primary prevention:** Avoid arthropod bites (repellents, protective clothing, tick checks) and practice safe handling of animals/carcasses (gloves). Vector-ecology interventions (e.g., prescribed fire affecting tick populations) are studied ([PMID: 40788927](#)).
- **Immunization: No licensed vaccine.** *"No licensed vaccine is available in the prophylaxis of tularemia and this is need of the time and high-priority research area"* ([PMID: 32989563](#)). The Live Vaccine Strain (LVS) remains investigational ([PMID: 42238599](#)).
- **Post-exposure prophylaxis (PEP): Doxycycline or ciprofloxacin (≈14 days) —** *"Streptomycin, gentamicin, doxycycline or ciprofloxacin are recommended for post-exposure prophylaxis"* ([PMID: 15677845](#)); CDC 2025 guidance details PEP after high-risk exposures ([PMID: 41026652](#)).
- **Public-health context:** *F. tularensis* is a **Tier-1 Select Agent / Category A bioterrorism agent** — *"Because F. tularensis has a low infectious inoculum, it is classified as a potential bioterrorism agent that could infect thousands of persons if intentionally released"* ([PMID: 41026652](#)); tularemia is a **notifiable disease** in the US.
- **Genetic screening / counseling:** Not applicable.

Section 14 — Other Species / Natural Disease

- **Taxonomy of hosts:** Broad — *"Tularemia is a bacterial disease of humans, wild, and domestic animals"* ([PMID: 32989563](#)). Natural reservoirs are **lagomorphs** (rabbits, hares)

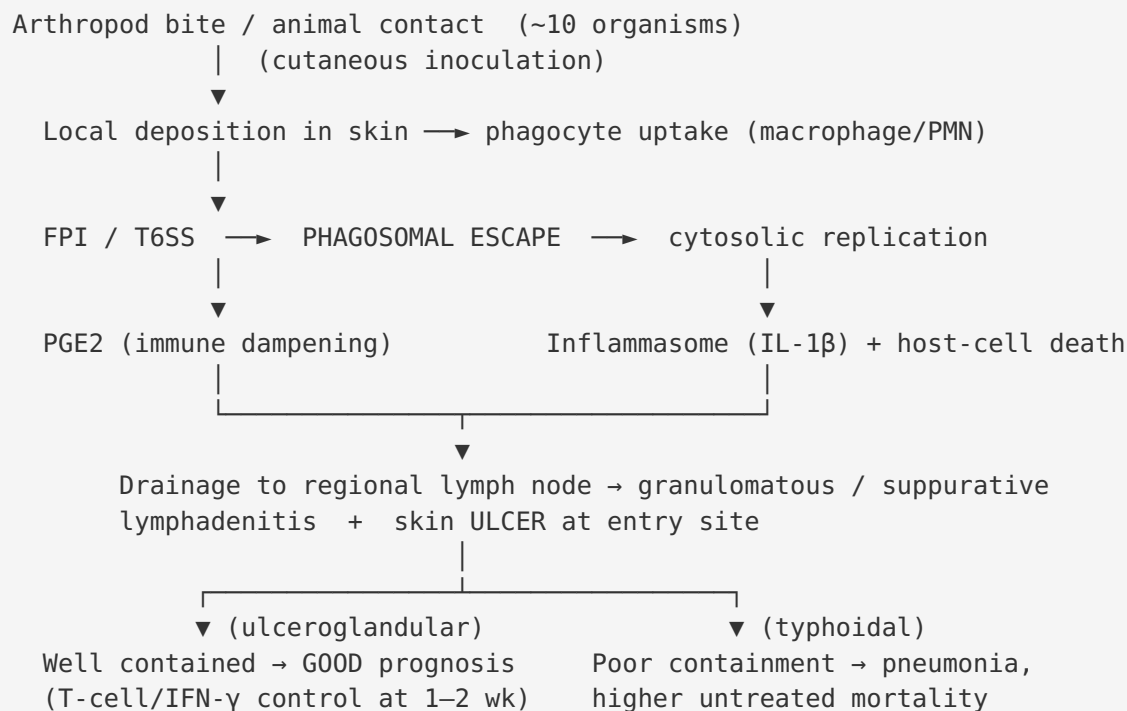
and **rodents**; arthropods (ticks, deer flies, mosquitoes) act as vectors/maintenance hosts. Tularemia is a Holarctic zoonosis ([PMID: 31600457](#)).

- **Veterinary relevance:** Causes natural disease and die-offs in lagomorphs and rodents ("rabbit fever"); cats and dogs may become infected and transmit to humans.
- **Comparative biology:** Intracellular pathogenesis (FPI/T6SS-dependent macrophage infection) is conserved across mammalian hosts; this conservation underlies the utility of animal models.
- **Zoonotic transmission:** Strongly zoonotic — humans are incidental hosts infected via vector bite, animal contact, ingestion, or inhalation. Human-to-human transmission does not occur.

Section 15 — Model Organisms

- **Principal model — mouse** (*Mus musculus*, NCBITaxon:10090): FPI/T6SS mutants (Δ iglB, Δ iglE, Δ pdpC) show "*incomplete phagosomal escape, and marked attenuation in the mouse model*" ([PMID: 23356941](#)) and "*showed marked attenuation in the mouse model*" ([PMID: 27830989](#)). C57BL/6 mice are used to assess FPI-mutant pathology in spleen/liver ([PMID: 27477000](#)). Mouse **inhalation models** are used to assess vaccine efficacy and identify transcriptional/translational biomarkers ([PMID: 35056485](#)).
- **In vitro models:** Macrophage cell lines **J774** and **THP-1**, and **human monocyte-derived macrophages (HMDM)** ([PMID: 27477000](#)); HeLa-FcγRII epithelial model for *F. novicida* ([PMID: 40748985](#)).
- **Surrogate strains:** *F. tularensis* LVS and *F. novicida* are widely used attenuated surrogates for mechanistic and vaccine studies ([PMID: 22493083](#)).
- **Phenotype recapitulation:** Mouse models faithfully reproduce intracellular replication, dissemination to spleen/liver, and lethality, and are the standard for vaccine-efficacy testing.
- **Limitations:** Attenuated surrogates (LVS, *F. novicida*) do not fully model virulent type A disease; localized ulceroglandular skin pathology is less emphasized than systemic/pulmonary endpoints in murine work.

Mechanistic Model / Interpretation



The unifying insight is that a **single molecular machine (the FPI-encoded T6SS)** converts *F. tularensis* from an ingested particle into a cytosolic replicating pathogen, and that the **balance between vigorous local cell-mediated immunity and bacterial containment** determines whether disease stays localized (ulceroglandular, good outcome) or disseminates (typhoidal/pneumonic, worse outcome). Because control is cell-mediated and antibody is comparatively unhelpful for killing, both diagnosis (serology at 2–3 weeks) and vaccine development (needing T-cell immunity) are shaped by this biology.

Evidence Base

PMID	Title (abbrev.)	Supports
38294108	<i>Systematic Review: Clinical Features, Treatment, Outcomes 1993–2023</i>	Ulceroglandular = most common form; treatment-specific fatality; low infectious dose
32989563	<i>Tularemia: a re-emerging tick-borne disease</i>	Clinical forms; transmission; antibiotics; no vaccine; zoonosis
3892222	<i>Tularemia: 30-year experience, 88 cases</i>	Incubation 3–6 d; two-syndrome framework; cell-mediated immunity; good prognosis
27830989	<i>IgE controls T6SS secretion</i>	FPI/T6SS → phagosomal escape → cytosolic replication; mouse attenuation
23356941	<i>LVS ΔpdpC phenotype</i>	Intramacrophage proliferation prerequisite; mouse-model attenuation
23403609	<i>Francisella mutants failing PGE2 induction</i>	FPI required for escape/replication; PGE2 immune modulation
27477000	<i>ΔpdpC/ΔiglG characterization</i>	HMDM/mouse models; spleen/liver pathology; PdpC regulation of iglABCD
16936340	<i>Tularemia in NW Turkey</i>	Late therapy → suppuration; PCR of node aspirates
29642835	<i>Oropharyngeal tularemia, E. Anatolia</i>	MAT $\geq$ 1/160 diagnostic threshold
39736154	<i>Tularemia — US, 2011–2022</i>	US incidence, geography, age/sex, race disparities
24280916	<i>Tularemia — US, 2001–2010</i>	Earlier surveillance; bimodal age/sex
34990926	<i>Cervical lymphadenopathy in children after tick bite</i>	Pediatric ulceroglandular; animal-handling route in Europe
42238599	<i>Tularemia & vaccination T-cell responses</i>	LVS/natural infection → multifunctional T cells
15677845 / 29183485	<i>Bichat guidelines</i>	First-line aminoglycosides; PEP with doxycycline/ciprofloxacin

PMID	Title (abbrev.)	Supports
41026652	<i>CDC 2025 treatment/prophylaxis</i>	Bioterrorism classification; treatment & PEP recommendations
40788927	<i>Prescribed fire & tick vectors</i>	Principal US tick vectors (<i>D. variabilis</i> , <i>A. americanum</i>)
22493083	<i>Perforin/granzyme protection</i>	Cytotoxic effector immunity in vaccine protection (model)
35056485	<i>Vaccine biomarkers, mouse inhalation</i>	Mouse inhalation model for vaccine efficacy
29761532	<i>Dermatological aspects, 168 cases</i>	Secondary skin manifestations; form distribution
31600457	<i>Ecology of Francisella</i>	Holarctic zoonosis; reservoir ecology

Limitations and Knowledge Gaps

- 1. Not a genetic disease.** Sections on host causal genes, pathogenic variants, inheritance, penetrance, epigenetics, and genetic testing are **not applicable**; the report reframes "molecular" content around bacterial virulence.
- 2. Form-specific data are sparse.** Much clinical data aggregate *all* tularemia; some cited series (Turkey) are oropharyngeal-predominant, so treatment/outcome numbers are partly extrapolated to the ulceroglandular form.
- 3. Quality-of-life data absent.** No EQ-5D/SF-36/PROMIS instruments have been applied specifically to ulceroglandular tularemia.
- 4. Human mechanistic evidence is indirect.** Core pathogenesis (FPI/T6SS, phagosomal escape) rests on cell-culture and mouse studies; direct human histopathological causal steps are inferred.
- 5. Diagnostic thresholds vary** (MAT $\geq 1/160$ vs $\geq 1:640$) across labs/outbreaks.
- 6. Vaccine gap.** No licensed vaccine; correlates of protective immunity in humans remain incompletely defined.

Proposed Follow-up Experiments / Actions

1. **Form-stratified outcome analysis** — mine the 870-case systematic-review dataset for ulceroglandular-only fatality, suppuration rate, and time-to-treatment thresholds.
 2. **Prospective PROs** — apply EQ-5D/SF-36 in an endemic-region cohort to quantify acute and residual QoL impact of ulceroglandular disease.
 3. **Time-to-antibiotic study** — formally test whether treatment within a defined window (e.g., ≤ 7 days) prevents lymph-node suppuration.
 4. **Host-immunogenetics** — although not Mendelian, explore whether host innate-immune polymorphisms (inflammasome, IFN- γ pathway) modulate ulceroglandular severity via candidate-gene/GWAS approaches.
 5. **Vaccine correlates** — leverage multifunctional T-cell signatures ([PMID: 42238599](#)) and mouse-model biomarkers ([PMID: 35056485](#)) to define human protective correlates for LVS-successor vaccines.
 6. **Skin-model development** — an intradermal small-animal model recapitulating the ulcer-plus-node phenotype would fill the gap left by systemic/pulmonary-focused mouse work.
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Report compiled from 10 confirmed findings across 5 investigation iterations and 38 reviewed papers. Evidence types: human clinical (case series, systematic reviews, national surveillance), model organism (mouse), and in vitro (macrophage cell lines/HMDM).
